Engineering-Prior-Guided NSGA-II for Differentiated Capacity-Load Ratio Optimization in County-Level 110 kV Grids with High Distributed PV Penetration

1st Dunbin Xu
State Grid Xuzhou Power Supply Company
Xuzhou, China
xzam@163.com

2nd Ke Wang
State Grid Xuzhou Power Supply Company
Xuzhou, China
17851199257@139.com

3rd Yan Xu*
State Grid Jiangsu Electric Power Design & Consulting Co., Ltd. Xuzhou Survey and Design Branch
Xuzhou, China
xuyan9999@163.com
*Corresponding Author

4th Ziyue Huang
State Grid Jiangsu Electric Power Design & Consulting Co., Ltd. Xuzhou Survey and Design Branch
Xuzhou, China
792572118@qq.com

5th Fei Geng
State Grid Jiangsu Electric Power Design & Consulting Co., Ltd. Xuzhou Survey and Design Branch
Xuzhou, China
2362324393@qq.com

6th Rui Yuan
State Grid Jiangsu Electric Power Design & Consulting Co., Ltd. Xuzhou Survey and Design Branch
Xuzhou, China
450532913@qq.com

***Abstract*—High penetration of distributed photovoltaic (PV) systems causes reverse power flow and affects N-1 supply adequacy in county-level 110 kV grids. This study jointly optimizes substation-specific capacity-load ratios (CLRs) and storage power ratings to minimize annualized cost and expected energy not supplied (EENS). The proposed Engineering-Prior-Guided NSGA-II (EPG-NSGA-II) combines engineering-prior-based population seeding with feasibility repair. On a county-level test system, EPG-NSGA-II identifies an elastic CLR design that reduces annualized cost by 3.4% and modeled EENS to zero relative to the 2.0-capped CLR design. EPG-NSGA-II achieves a 4.8% higher mean final hypervolume than standard NSGA-II for the 40-substation case.**

Keywords—capacity-load ratio, penetration of distributed photovoltaic, multiobjective optimization, reverse power flow, 110 kV grid planning.

I. Introduction

High penetration of distributed PV causes midday export and reverse power flow at 110 kV substations [1], potentially affecting supply adequacy under N-1 contingencies. A fixed CLR cap of 2.0 cannot reflect substation-level differences in PV penetration and transfer capability [2]. Multiobjective evolutionary algorithms approximate cost–adequacy trade-offs [3], while knowledge-transfer strategies improve search efficiency [4], [5]. However, neither explicitly handles the joint CLR and reverse-flow constraints in this planning problem, so infeasible candidates remain common.

EPG-NSGA-II addresses this gap by augmenting standard NSGA-II with engineering-prior-based population seeding and feasibility repair. The bi-objective model jointly optimizes substation-specific CLRs and storage power ratings subject to aggregate-CLR and reverse-flow constraints, minimizing annualized cost and EENS. EENS quantifies supply inadequacy under N-1 contingencies rather than imposing a hard N-1 constraint. Comparative and ablation studies use a county-level test system constructed from publicly available data.

II. Engineering-Prior-Guided NSGA-II

A. Planning Model

The county-level grid contains K 110 kV substations. At station j, R_j is the CLR (planned transformer capacity over forecast peak load), P_j the storage power rating, and L_j^p the five-year forecast peak load using a 5% annual growth planning assumption. Station CLRs obey the load-weighted aggregate band.

At station j, pre-storage midday reverse injection is r_j^0; after storage absorption, residual reverse flow is

$$\tilde{r}_j(P_j)=\left[r_j^0-P_j\right]^+. \tag{1}$$

Here, $[x]^+=\max\{0,x\}$. With $R_L=1.8$ and $R_U=3.0$, the load-weighted aggregate CLR is constrained as

$$R_L\le\frac{\sum_{j=1}^{K}R_jL_j^p}{\sum_{j=1}^{K}L_j^p}\le R_U. \tag{2}$$

With $\beta=0.85$, $R_{220}=1.8$, $\delta=0.85$, and $\eta=1.0$ denoting the reverse-loading limit, upper-level CLR, load-coincidence factor, and reverse-power coincidence factor, respectively, the station and optional upstream reverse-flow constraints are

$$\begin{gathered}\tilde{r}_j(P_j)\le\beta R_jL_j^p,\\ \eta\sum_{j=1}^{K}\tilde{r}_j(P_j)\le\beta R_{220}\delta\sum_{j=1}^{K}L_j^p.\end{gathered} \tag{3}$$

Annualized cost f_1 and EENS f_2 are minimized as

$$\begin{gathered}\min(f_1,f_2),\\ f_1=\sum_{j=1}^{K}c_RR_jL_j^p+c_PP_j+C_j^{loss}+C_j^{om},\\ f_2=\sum_{j=1}^{K}EENS_j.\end{gathered} \tag{4}$$

Here, c_R and c_P are unit costs; C_j^{loss} and C_j^{om} are annual loss and O&M costs. For outage state $s\in\Omega_j$ at substation j, p_s, h_s, and $S_{j,s}$ denote occurrence probability, duration, and available supply, respectively; the resulting EENS is

$$EENS_j = \sum_{s \in \Omega_j} p_s \, h_s \max\{0, L_j^p - S_{j,s}\}. \quad (5)$$

B. EPG-NSGA-II Procedure

EPG-NSGA-II retains standard NSGA-II variation and selection [3]. Algorithm 1 applies common-offset seeding plus station-level and optional upstream repair; the same repair is applied after variation.

Algorithm 1 Engineering-Prior-Based Seeding and Feasibility Repair

Input: population size N; K substations; bounds $R_{min} \le R_{i,j} \le R_{max}$ and $0 \le P_{i,j} \le P_{max}$. Indices i and j denote individuals and substations, respectively; $R_{i,j} \sim U([R_{min}, R_{max}])$ denotes uniform sampling; clip applies the stated bounds; ← denotes assignment.
Output: N repaired candidates.

```
for i=1,…,N do
    if constructing the initial population then
        Sample R_{i,j} ~ U([R_min, R_max]) and set P_{i,j} = 0 for all j
    else
        Retain the post-variation offspring values
    end if
    Apply a bounded common offset to all R_{i,j} until (2) holds
    Update P_{i,j} using the station-level repair rule:
        P_{i,j} ← clip( max{ P_{i,j} [r_j^0 - β R_{i,j} L_j^p]^+ }, 0, P_max )
    Order substations by descending residual reverse injection
    if the upstream constraint is enabled then
        Increase adjustable P_{i,j} until the upstream inequality in (3) holds or all
        adjustable ratings reach P_max
    end if
    Save R_{i,j} and P_{i,j} as candidate i
end for
return N repaired candidates
```

The common-offset adjustment is defined as

$$R_{i,j} \leftarrow \mathrm{clip}\left(R_{i,j} + \lambda_i, R_{min}, R_{max}\right), \quad (6)$$

where λ_i enforces the load-weighted aggregate CLR in (2). After station-level storage repair, the upstream violation is

$$v_{up}(x_i) = \left[\eta \sum_{j=1}^{K} \tilde{r}_j(P_{i,j}) - \beta R_{220} \delta \sum_{j=1}^{K} L_j^p\right]^+ . \quad (7)$$

When $v_{up} > 0$, storage is allocated in descending residual-injection order until $v_{up} = 0$ or adjustable ratings reach $P_{\max}$. Residual infeasibility is handled by constraint domination. These operators modify only candidate construction and repair; standard NSGA-II ranking and environmental selection remain unchanged.

III. Experiments

A. Setup

Station-level loads are generated by scaling the IEEE 33-bus load data [6], and PV profiles are derived from PVGIS irradiance [7]. The population size is $N = 100$, with three independent runs. Standard NSGA-II and NSGA-III use an 800-generation reference budget; EPG-NSGA-II and its ablations use 400 generations.

B. Efficiency and Ablation

TABLE I. Normalized Mean Final Hypervolume (HV)

Method	K=10	K=20	K=30	K=40
NSGA-II	1.000	1.000	1.000	1.000
NSGA-III	0.998	0.991	0.979	0.985
Seeding-only	0.996	0.991	0.988	0.998
Repair-only	1.004	1.006	1.008	1.048
EPG-NSGA-II	1.007	1.007	1.009	1.048

Table I reports mean final HV at each method's termination, normalized by standard NSGA-II for each K using a common reference point. EPG-NSGA-II matches or exceeds the baseline, yielding a 4.8% improvement at K = 40. Repair-only nearly matches the full method, whereas seeding-only remains close to the baseline. At $K = 40$, the mean EPG-NSGA-II curve reaches the standard NSGA-II 800-generation final HV by generation 84.

C. CLR Comparison at K = 20

Using the same knee-point selection rule for both Pareto sets, the elastic CLR design reduces annualized cost from 236.40 to 228.43 million CNY/yr and modeled EENS from 561 to 0 MWh/yr relative to the 2.0-capped CLR design. The source-load ratio, defined as installed PV capacity divided by current peak load, is positively associated with the selected CLR, while storage is used as a supplement (Fig. 1).

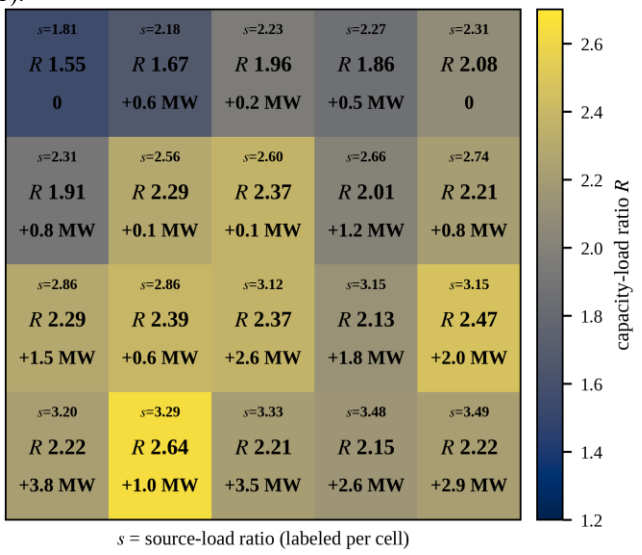


Fig. 1. Differentiated CLRs and storage ratings for the $K = 20$ case.

IV. Conclusion

EPG-NSGA-II attains a 4.8% higher mean final HV than standard NSGA-II at $K = 40$, while the elastic CLR design reduces annualized cost by 3.4% and modeled EENS to zero under the current model relative to the 2.0-capped CLR design. Future work will add storage energy/SOC limits and real-grid calibration.

References

[1] Q. Hou, N. Zhang, E. Du, M. Miao, F. Peng, and C. Kang, "Probabilistic duck curve in high PV penetration power system: Concept, modeling, and empirical analysis in China," Applied Energy, vol. 242, pp. 205–215, 2019, doi: 10.1016/j.apenergy.2019.03.067.

[2] J. Xiao, R. Liu, and M. Long, "Relationship between substation capacity-load ratio and the next-level medium-voltage distribution network," Electric Power Construction, vol. 36, no. 11, pp. 45–50, 2015 (in Chinese).

[3] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," IEEE Trans. Evol. Comput., vol. 6, no. 2, pp. 182–197, 2002.

[4] X. Xue, C. Yang, Y. Hu, K. Zhang, Y.-M. Cheung, L. Song, and K. C. Tan, "Evolutionary sequential transfer optimization for objective-heterogeneous problems," IEEE Trans. Evol. Comput., vol. 26, no. 6, pp. 1424–1438, 2022.

[5] X. Xue, Y. Hu, L. Feng, K. Zhang, L. Song, and K. C. Tan, "Surrogate-assisted search with competitive knowledge transfer for expensive optimization," IEEE Trans. Evol. Comput., vol. 29, no. 6, pp. 2416–2430, 2025.

[6] M. E. Baran and F. F. Wu, "Network reconfiguration in distribution systems for loss reduction and load balancing," IEEE Trans. Power Del., vol. 4, no. 2, pp. 1401–1407, 1989.

[7] European Commission, Joint Research Centre, "Photovoltaic Geographical Information System (PVGIS)." [Online]. Available: https://re.jrc.ec.europa.eu/pvg_tools/en/.